

A Compact Bath-Temperature Extension of the LTE Hammer–Rosen Marshak-Wave Model

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Abstract

The original LTE Hammer–Rosen (HR) theory assumes that the surface temperature $T_s(t)$ is given. Here we summarize a compact bath-temperature version in which the input is the incident bath temperature $T_b(t)$ and the surface temperature is estimated from a Marshak boundary condition. The model keeps the original HR front and profile structure, avoids differentiating the noisy bath history, and requires only fluence integrals plus one scalar algebraic solve at each time.

1 LTE diffusion equation and material model

The LTE material model is

$$e = fT^\beta \rho^{-\mu}, \quad \frac{1}{\kappa_R} = gT^\alpha \rho^{-\lambda}, \quad (1)$$

where e is the specific material energy, κ_R is the Rosseland opacity, ρ is constant, and $f, \beta, \mu, g, \alpha, \lambda$ are material constants. The LTE diffusion equation can be written as

$$\boxed{\frac{\partial T^\beta}{\partial t} = C \frac{\partial^2 T^n}{\partial x^2}} \quad (2)$$

with

$$\boxed{n = 4 + \alpha, \quad \epsilon = \frac{\beta}{n}} \quad (3)$$

and

$$\boxed{C = \frac{16g\sigma}{3nf} \rho^{-2+\mu-\lambda}.} \quad (4)$$

Here σ is the Stefan–Boltzmann constant in the unit system used for the HR coefficients.

2 Original surface-temperature HR formulas

For a prescribed surface temperature $T_s(t)$, define

$$H(t) = T_s^n(t). \quad (5)$$

The first-order HR front formula is

$$\boxed{x_F^2(t) = K_\epsilon C H^{-\epsilon}(t) \int_0^t H(t') dt'} \quad (6)$$

where

$$K_\epsilon = \frac{2 + \epsilon}{1 - \epsilon}. \quad (7)$$

Equivalently, since $H^{-\epsilon} = T_s^{-\beta}$,

$$x_F^2(t) = K_\epsilon C T_s^{-\beta}(t) \int_0^t T_s^n(t') dt'. \quad (8)$$

The HR material energy per unit area is

$$E(t) = f \rho^{1-\mu} (1 - \epsilon) x_F(t) H^\epsilon(t). \quad (9)$$

Combining (6) and (9) gives

$$E^2(t) = Z_1 H^\epsilon(t) \int_0^t H(t') dt' \quad (10)$$

with

$$Z_1 = f^2 \rho^{2(1-\mu)} (2 + \epsilon) (1 - \epsilon) C. \quad (11)$$

3 Bath boundary condition

The bath-driven problem prescribes $T_b(t)$ rather than $T_s(t)$. We use the Marshak boundary condition in the convention

$$F(t) = 2\sigma [T_b^4(t) - T_s^4(t)]. \quad (12)$$

Energy conservation gives

$$E(t) = \int_0^t F(t') dt'. \quad (13)$$

The exact Cohen–Heizler boundary-temperature method solves (10)–(13) by time marching. The compact model below replaces that marching calculation by a fluence approximation.

4 Fluence-ratio approximation

Define the instantaneous surface-to-bath ratio

$$u(t) = \left[\frac{T_s(t)}{T_b(t)} \right]^4, \quad (14)$$

so that

$$T_s(t) = T_b(t) u^{1/4}(t). \quad (15)$$

The approximation is that this ratio is slowly varying over the accumulated fluence history:

$$\frac{T_s(t')}{T_b(t')} \approx \frac{T_s(t)}{T_b(t)}, \quad 0 < t' < t. \quad (16)$$

Therefore

$$\int_0^t T_s^4(t') dt' \approx u(t) \int_0^t T_b^4(t') dt', \quad (17)$$

and

$$\int_0^t T_s^n(t') dt' \approx u^{n/4}(t) \int_0^t T_b^n(t') dt'. \quad (18)$$

Define the two bath fluences

$$J_4(t) = \int_0^t T_b^4(t') dt', \quad J_n(t) = \int_0^t T_b^n(t') dt'. \quad (19)$$

Using (12), (13), and the fluence approximation gives

$$E(t) \approx 2\sigma J_4(t) [1 - u(t)]. \quad (20)$$

Using (10) gives

$$E^2(t) \approx Z_1 u^{(n+\beta)/4}(t) T_b^\beta(t) J_n(t), \quad (21)$$

where $n\epsilon = \beta$ has been used. Equating (20) and (21) yields the scalar algebraic equation for $u(t)$:

$$1 - u(t) = \left[\frac{Z_1 T_b^\beta(t) J_n(t)}{[2\sigma J_4(t)]^2} \right]^{1/2} u^{(n+\beta)/8}(t). \quad (22)$$

This equation is the only extra operation relative to the original surface-HR formula.

5 Bath-driven front formula

Once $u(t)$ is known, the bath-driven front is

$$x_F^2(t) = K_\epsilon C u^{(n-\beta)/4}(t) T_b^{-\beta}(t) J_n(t). \quad (23)$$

Equivalently,

$$x_F(t) = \left[K_\epsilon C u^{(n-\beta)/4}(t) T_b^{-\beta}(t) J_n(t) \right]^{1/2}. \quad (24)$$

If $u \rightarrow 1$, then $T_s \rightarrow T_b$ and (23) reduces to the naive surface-HR formula with $T_s = T_b$.

6 Temperature profile

After computing $u(t)$ and $x_F(t)$, set

$$T_s(t) = T_b(t) u^{1/4}(t), \quad H(t) = T_s^n(t). \quad (25)$$

Define

$$y = \frac{x}{x_F(t)}. \quad (26)$$

The first-order HR profile is

$$T(x, t) = T_s(t) \left[(1 - y) \left(1 + \frac{\epsilon}{2} (1 - A) y \right) \right]^{1/[n(1-\epsilon)]} \quad (27)$$

for $0 \leq y \leq 1$, and $T = 0$ for $y > 1$. The profile parameter is

$$A(t) = \frac{x_F^2(t)}{C H^{1-\epsilon}(t)} \frac{\dot{H}(t)}{H(t)}. \quad (28)$$

For a leading-order profile, use

$$T(x, t) \approx T_s(t) \left(1 - \frac{x}{x_F(t)} \right)^{1/n}. \quad (29)$$

7 Numerical algorithm

Given tabulated $T_b(t)$:

1. Interpolate $T_b(t)$ on a convenient grid. For noisy data, linear interpolation is usually safest.
2. Compute $J_4(t_i)$ and $J_n(t_i)$ by trapezoidal integration.
3. For each time t_i , if $T_{b,i} = 0$ or either fluence is zero, set $T_{s,i} = 0$ and $x_{F,i} = 0$.
4. Otherwise solve (22) for $u_i \in (0, 1)$. Define

$$\Phi(u; t_i) = 1 - u - \left[\frac{Z_1 T_{b,i}^\beta J_{n,i}}{(2\sigma J_{4,i})^2} \right]^{1/2} u^{(n+\beta)/8}. \quad (30)$$

Since $\Phi(0; t_i) > 0$, $\Phi(1; t_i) < 0$, and $\partial\Phi/\partial u < 0$, there is a unique physical root. Bisection or Brent on $(0, 1)$ is robust.

5. Compute $T_{s,i} = T_{b,i} u_i^{1/4}$.
6. Compute $x_{F,i}$ from (24).
7. For profiles, compute $H_i = T_{s,i}^n$, estimate $\dot{H}_i$ by finite differences, compute A_i from (28), and use (27).

8 Self-contained summary

The bath-driven LTE HR model solves the diffusion equation

$$\frac{\partial T^\beta}{\partial t} = C \frac{\partial^2 T^n}{\partial x^2}, \quad n = 4 + \alpha, \quad \epsilon = \frac{\beta}{n}, \quad (31)$$

with the material model

$$e = f T^\beta \rho^{-\mu}, \quad \frac{1}{\kappa_R} = g T^\alpha \rho^{-\lambda}, \quad (32)$$

and the bath boundary condition

$$F = 2\sigma(T_b^4 - T_s^4). \quad (33)$$

Given $T_b(t)$, compute

$$J_4 = \int_0^t T_b^4(t') dt', \quad J_n = \int_0^t T_b^n(t') dt'. \quad (34)$$

Then solve

$$1 - u = \left[\frac{Z_1 T_b^\beta J_n}{(2\sigma J_4)^2} \right]^{1/2} u^{(n+\beta)/8}, \quad Z_1 = f^2 \rho^{2(1-\mu)} (2 + \epsilon) (1 - \epsilon) C. \quad (35)$$

The surface temperature is

$$T_s = T_b u^{1/4}, \quad (36)$$

and the front is

$$x_F^2 = \frac{2 + \epsilon}{1 - \epsilon} C u^{(n-\beta)/4} T_b^{-\beta} \int_0^t T_b^n(t') dt'. \quad (37)$$

The profile is

$$T(x, t) = T_s(t) \left[(1 - y) \left(1 + \frac{\epsilon}{2} (1 - A)y \right) \right]^{1/[n(1-\epsilon)]}, \quad y = \frac{x}{x_F(t)}, \quad (38)$$

with

$$A = \frac{x_F^2}{C T_s^{n(1-\epsilon)}} \frac{d}{dt} \ln T_s^n. \quad (39)$$

The HR spatial approximation has the same formal order as the original Hammer–Rosen theory: the retained front and profile are first order in ϵ , with neglected HR spatial terms of order $O(\epsilon^2)$. The additional bath approximation is the fluence-ratio assumption

$$T_s(t')/T_b(t') \approx T_s(t)/T_b(t) \quad (0 < t' < t). \quad (40)$$

Thus the model is most accurate when the surface-to-bath ratio varies smoothly over the accumulated fluence history. It is not the exact Cohen–Heizler boundary evolution, but it is stable, compact, and avoids differentiating the bath drive.